

Vision-Based Navigation for AMRs in a Dynamic Warehouse

Progress summary for committee review

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1. The question

Can a vision-based pipeline provide reliable localization for an autonomous mobile robot in a warehouse that contains moving objects? The work compares two stereo-inertial front ends, ORB-SLAM3 and VINS-Fusion, under controlled changes in scene dynamics, floor texture and compute load, and asks whether masking dynamic objects with a learned detector improves localization. Path planning is outside the scope: the deliverable is a reliable navigation state and map, not a planner.

2. What exists

A warehouse simulator with static, dynamic, textured-floor and featureless-floor variants; both estimators instrumented to record per-frame timing, delivery, state, CPU and memory; a custom YOLO detector feeding dynamic-object masks into both; RTAB-Map as a separate mapping and loop-closure back end; and a run-logging and validity-audit harness. The evidence base is 66 logged estimator runs across seven simulation datasets at three repeats per condition, five multi-sensor recordings spanning four worlds, 9860 camera frames with projected object ground truth, and one reconstructed map with geometry and drivability evaluation.

3. What is established

A wheel+IMU EKF currently outperforms both visual pipelines in static scenes. On five recordings spanning four worlds, the EKF achieves the lowest aligned ATE on four of five, at 0.10 % to 0.88 % drift.

Dataset	IMU only	Wheel odom.	VINS-Fusion	ORB-SLAM3	EKF
allsensor_000	21.70	2.80	0.62	0.58	0.19
allsensor_001	12.62	0.98	0.38	0.37	0.31
allsensor_002	13.24	1.49	0.56	0.60	0.26
allsensor_003	45.56	3.19	0.78	6.48	0.42
allsensor_004	26.71	3.24	0.43	1.24	0.49

Aligned ATE RMSE in metres, rigid $SE(3)$ fit with scale fixed.

This is the baseline the project previously lacked: until now every trajectory figure reported visual error with nothing to compare it against. The EKF’s advantage is specific to these conditions — static scene, simulated wheel encoders, no slip — so it is a floor the visual system must clear rather than a replacement for it. It does, however, sharpen the thesis question: vision must justify itself where dead reckoning fails, which means long runs, wheel slip, relocalization after tracking loss, and moving obstacles.

The detector’s limit is detection, not semantics. Over 9860 frames the detector reaches precision 0.684 and moving-object recall 0.591, while class assignment is 0.996 accurate once an object has been localized. Improving the classifier would change nothing; recall is the binding constraint. The model also produces bounding boxes rather than segmentation, and class membership is not evidence of motion.

The reconstructed map is geometrically sound. Against the simulator’s own collision geometry at 0.25 m tolerance, occupied-cell precision is 96.6 % with 89.5 % recall over the region the robot observed. The lower whole-building figure of 67.2 % is route coverage — the run never entered the side aisles — not mapping failure.

Neither pipeline runs at input rate. Frames arrive every 33 ms; ORB-SLAM3 tracking takes 63 ms and its local mapping 190 ms, and enabling the detector raises these to 113 and 356 ms. Input retention falls from 89 % to 67 %.

4. What is not established

The dynamic-scene results are withheld. Those worlds are not yet a physical simulation: props follow scripted kinematic paths, carrying collision geometry but producing no dynamic response against the robot. Forty-eight runs were logged there and all forty-eight fail the plausibility check, but those numbers describe the simulation rather than the problem, so they are excluded here and retained in the record as excluded evidence. The consequence is that the condition this thesis is named for has *no valid testbed* — untested, rather than tested and failed. Rebuilding those worlds is the first item of the next period and the cheapest candidate explanation for the breakdown itself.

The core hypothesis has no valid test behind it. Two instrumentation faults prevent it, and the environment question above compounds them. The VINS-Fusion filter never fired: all twelve runs requested it and all twelve record zero matched detector frames and zero removed tracks, even though the detector works offline. Separately, eleven of twelve ORB-SLAM3 baseline/filtered pairs received frame counts differing by 10 % to 30 % on the same recording. One admissible pair exists out of twenty-four. No cross-treatment accuracy conclusion is drawn. Both are code-path faults and stand regardless of scene, but the filtered runs were recorded in the dynamic worlds, so the hypothesis now needs a working mask path *and* a valid environment.

5. How the work is validated

Every comparison is tested for admissibility before its numbers are read: the same recorded input to both treatments within 5 %, the intended treatment demonstrably active rather than merely requested, a common ground-truth interval with the matched sample count reported, and physical plausibility. Four claims were retracted or prevented this period by that protocol — a camera-rotation parsing bug found through map distortion while ATE looked plausible throughout; a map wrongly judged unnavigable because its grid was mis-seated; a drivability score discovered to have been measured against the map’s own construction run; and the two filter faults above. Each would otherwise have entered the thesis as a result.

6. Next period, in priority order

1. Repair the VINS mask delivery path. This alone unblocks the core hypothesis, and needs no additional data.
2. Equalise ORB-SLAM3 input across paired runs so the filter comparison becomes admissible.
3. Rebuild the dynamic worlds with physically simulated props, giving the titular condition a valid testbed.
4. Add disturbance to the simulation — wheel slip and encoder quantisation — since both idealisations currently favour the dead-reckoning baseline.
5. Record an independent route for map drivability, and extend the map evaluation to the large warehouse.

7. Guidance requested

1. **Scope.** Given that a wheel+IMU EKF currently beats both visual pipelines in static scenes, should the thesis argue for vision as the primary localization sensor, or reframe around the conditions under which vision adds value over dead reckoning?
2. **Depth or breadth.** Rebuild the dynamic environment properly before measuring anything further in it, or extend to the real robot and the large warehouse with the static results as they stand?
3. **Integration.** Relocalization, global optimization and multi-session map merging are listed as future work. Is that the right next contribution, or premature until dynamic-scene tracking is sound?

Full detail, including every retracted, excluded and failed measurement, is in the accompanying progress report. Results flagged there in red are measured but not yet trustworthy as evidence, and are excluded from the claims above.